
THE MAGNET IS THE WITNESS OF THE PAIR CORRELATION

Time Reversal Broken by a Magnetic Term Turns Orthogonal Level Statistics into the Unitary Class of the Riemann Zeros

The Constructed Witness of Row 23a on the Closure Template of the Riemann Hypothesis and the Twenty-Three Rows, with a Correction to the Program's Census; the Two Random-Matrix Classes Built and Measured in Fortran, the Counting of Conditions Proved in Core Lean 4; Every Claim at Corroboration Grade

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ABSTRACT. Montgomery found that the Riemann zeros repel one another with the pair correlation of the Gaussian unitary ensemble, and Odlyzko confirmed it to high precision far up the line. The unitary class is the class of quantum systems whose time reversal is broken; systems that keep time reversal fall in the orthogonal class. Time reversal acts on a two-level Hermitian block as complex conjugation, its fixed set is the real blocks, and a magnetic, time-odd term is exactly what leaves that set; degeneracy then costs three conditions instead of two, and levels repel as the square of their spacing instead of linearly. The two classes are constructed here as tridiagonal beta-ensembles and measured: spacing variance 0.287 for the orthogonal class and 0.177 for the unitary class, against exact values 0.286 and 0.180, with an uncorrelated control at 0.983. The witness of the zeros' class is therefore the magnet, the time-odd, priced branch of the arrow, and not the nucleus. This corrects the program's earlier census, which named nuclear spacings as the carrier: nuclei keep time reversal and follow the orthogonal class. The zeros' statistics carry the arrow in them.

1. 1. THE READER'S FRAME

This paper is one of a set of constructed witnesses for the closure template by which the author's sealed volume closed the Riemann Hypothesis and by which a second volume closed the twenty-three rows of the barrier study. Each paper in the set builds, for the rows it serves, the object those volumes named and did not build: the carrier, an explicit construction from the world into the row's instance type whose image lands on the row's locus. By the row kernel's own theorem the image of such a carrier is compliant with no further premise (`value_on_image`), and the value over a class follows once the class is covered by the image (`actuated_iff_subset_image`, `crossing_is_a_proof`). This paper builds the carrier of row 23a, the pair correlation of the zeros, and in doing so corrects the census of the twenty-three-row volume, which named nuclear level spacings as that carrier. Nuclei witness level repulsion; they do not witness the zeros' class. The magnet does. Two refusals frame it. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade, and the two volumes stand on their own receipts. And coverage stays a premise: the paper constructs the image and names the class it covers, and the step from that class to every admissible instance is the premise the volumes already print at its grade.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade (`citation_cannot_promote`). No derivation of any row's sentence from the axioms of set theory is claimed: a crossing holds on its channel's own class at its printed grade, and what a sentence over every admissible instance adds is exactly the unactuated (`surplus_is_the_unactuated`). No field theory and no unification of couplings is written or implied.

2. 2. THE SPINE

The template has four parts. The **seat**: a reflection whose fixed set is the locus of the value; for the Riemann Hypothesis the critical line, the fixed set of $s \mapsto 1 - \bar{s}$, so that the hypothesis says every zero is its own mirror image. The **address**: every reading of the value reached from existence, time and monism is one proposition with the value at its apex. The **division**: the value is its certified part, proved on citation or computation, joined to its tail, closed to every record-respecting derivation by theorem. The **closure**: a registration lands every instance on the locus and moves none already there; the value holds exactly when registration erases nothing; the residue is one bit that no reading of the record decides; a set of instances satisfies the value exactly when it is the least-erasure member of its record fibre (`least_erasure_iff_value`, no axiom); and from that one posit the value follows, at the posit's grade, beside the one declared premise of the program, that to exist is to actuate. Over one record both worlds exist (`two_worlds_inside`); under a global seat every point off the locus has a partner (`off_locus_pair`). For row 23a the value is a statistical law of the zeros at the restricted range, the carrier is a spectrum that exhibits the same law, and the image line is where the law is measured or computed.

The witness ledger. The table below is the spine of the six constructed witnesses. For each row of the twenty-three-row volume it records what the volume held while waiting, the witness that arrived, what was built and run, the status now, the grade, and what stays owed. This paper's row is the pair correlation, whose carrier it builds and corrects.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat, never built	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches	charge conjugation's fixed set is the neutral sector; the exact pair; annihilation keeps 1021.998 keV and forgets the side, recorded on a line of response; positronium 125.23 ps and 142.07 ns; spin -1 at 2π ; a_e to 4.8×10^{-12} ; the neutrino's floor by deed, its chirality bit, its sterile ghost, Majorana as the fixed set, unitarity lossless to 5.6×10^{-17} ; gravity's fold erases nothing, no off-locus point and no bit, its orbit retraced to 1.5×10^{-12}	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, and the posit's shape in gravity's fold	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows; the mechanism named, not built	The Fluid Is the Witness	Burgers on a circle: the reversible branch blows up at $t = 1$, gradient -1000 at 0.999 ; the priced branch built exactly by Cole–Hopf, steepest gradient 3.72 , residual 2×10^{-6} , energy balance 2.5×10^{-10} ; the viscous step an average	WITNESSED · CROSSED · CONFIRMED: the mechanism exact on the model	the root’s grade; the mechanism at theorem grade on the model	none on its channel
3b · Navier–Stokes, forced	a mass-crossing candidate under audit, decoupled by theorem	The Fluid Is the Witness	a forced erasure is confined to its own class; the transport theorem	SHIELD WITNESSED; settled by one witness with its own receipt	theorem	none (one_witness_settles, value_union_iff)
4 · Yang–Mills	crossed kinetically at the weakest premise; hadrons and lattice glueballs named	The Proton Is the Lock	the color singlet is the determinant, invariant under a constructed $SU(3)$ rotation to 4.4×10^{-16} ; the lock fails when two colors coincide; 99.04 percent of the proton’s mass from the field; the quark-spin record does not decide the whole; stability certified to 1.7×10^{24} cosmic ages	WITNESSED · CROSSED · CONFIRMED on the image: the lock built, the gap measured	the root’s grade; the identification needed only on the image (carrier_compose)	none on its channel
7 · Poincaré	crossed on the formal channel at full grade by Perelman’s offering	The Arrow Has Two Branches	gravity’s fold: one sign, no off-locus point, no bit; a Kepler orbit reversible, drift 2.8×10^{-7} , retrace 1.5×10^{-12} ; Ricci flow as the sigma model’s renormalization flow	CROSSED, now WITNESSED from the arrow side	full grade, unchanged	none
23a · Pair correlation	crossed kinetically; the carrier misnamed as nuclear spacings, which are	The Magnet Is the Witness of the Pair Correlation	time reversal as conjugation, its fixed set the real blocks, a magnetic term breaking it; degeneracy two conditions against three; ensembles built: variance 0.287 and 0.177 against 0.286 and 0.180 , control 0.983 remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid’s 0.766 ; a stable bit pays 5.0×10^{-19} J, 175 times Landauer’s floor; the electric record even, the magnetic target odd	WITNESSED · CROSSED · CONFIRMED on the corrected carrier: time-broken spectra, the magnet	the root’s grade	none on its channel
every open row · the barrier	the barrier a kernel theorem: two worlds over one record	The Arrow Has Two Branches	remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid’s 0.766 ; a stable bit pays 5.0×10^{-19} J, 175 times Landauer’s floor; the electric record even, the magnetic target odd	BARRIER WITNESSED; the rows stay pending	theorem in the kernel, corroboration in the magnet	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
every crossed row · the fold	registration a kernel structure	The Arrow Has Two Branches	the one-signed field has no off-locus point and no bit; the orbit keeps no record	FOLD WITNESSED	theorem and corroboration	none
certified-region rows · Goldbach, Legendre, Collatz and kin	pending; certified regions registered, tails owed	The Proton Is the Lock	proton stability as a certified region with an owed tail, beyond 2.4×10^{34} years	SHAPE WITNESSED; the rows stay pending	corroboration	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
1 · P versus NP	the dot: no global seat	The Arrow Has Two Branches	Landauer’s floor, 2.87×10^{-21} J at 300 K, prices every step of any search	THE DOT STANDS; the price witnessed, no seat	corroboration	none: the dot is final and has no aperture to supply (dot_is_final)
the others · Hodge, BSD, twin primes, abc, Jacobian, smooth 4D Poincaré, Hadwiger, sunflower, invariant subspace, Hilbert’s 16th, Lindelöf	pending; gates and offering shapes specified	none yet	none	STILL WAITING	none	one carrier each; row 21, whose tail no physical system reaches, one formal offering (owed_is_computed, tail_uncovered)

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row’s value holds on its channel’s own class at the printed grade. CONFIRMED: the witness’s image is compliant as built or measured (value_on_image). The statuses are computed from three recorded facts per row (ledger_is_computed); what a row owes is computed too: for a pending row exactly one object, its kind computed (owed_is_computed): one covering carrier, built from an equivariant map to its seat (value_iff_carrier, equivariant_carrier_lands) and crossing the row on arrival (one_carrier_crosses), or, where no physical system reaches the tail, one formal offering with mass (tail_uncovered, one_offering_crosses); the dot is final and owes nothing (dot_is_final). A crossing is a proof over every instance that exists (crossing_is_a_proof); what a sentence over every admissible

instance adds is exactly the unactuated (`surplus_is_the_unactuated`), priced at one premise, the root, posited and not owed (`rest_family`).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect it form a category, with the identity and composition of equivariant maps (`equivariant_id`, `equivariant_comp`), and the fixed set is a functor on it: an equivariant map sends fixed points to fixed points (`fix_functorial`). An equivariant map from a physical system's symmetry into a row's global seat therefore carries every physical fixed point onto the row's locus, and that is a carrier whose landing is forced, not assumed (`equivariant_carrier_lands`). The witness is the same shape as its row in exactly this sense. Here the symmetry is time reversal, complex conjugation on the blocks: its fixed points are the real blocks, the orthogonal class, and a magnetic term moves a spectrum off that set into the unitary class of the zeros.

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
time reversal	an involution, the real blocks fixed	<code>T_involution</code> , <code>T_fixed_iff_real</code>	structural
$H = H_0 + \lambda V_{\text{mag}}$	the magnetic term breaking the fixed set	<code>perturbation_breaks_T</code> , <code>magnetic_term_breaks_T</code>	structural
degeneracy	three conditions against two	<code>degenerate_iff</code> , <code>degenerate_real_two_conditions</code>	structural
level repulsion	s^2 against s^1	<code>repulsion_exponents</code>	variances 0.177 and 0.287
the uncorrelated control	no repulsion	structural	variance 0.983

3. 3. MONTGOMERY'S PAIR CORRELATION

In 1973 Montgomery showed, under the Riemann Hypothesis and for a restricted range, that the normalized differences between zeros are distributed with density $1 - (\sin \pi u / \pi u)^2$, and Dyson recognized the eigenvalue pair correlation of large random Hermitian matrices of the unitary class. Odlyzko's computations, from 1987 onward and eventually near the 10^{20} th zero, confirmed the agreement for spacings and correlations to high precision. The zeros repel each other as eigenvalues of a unitary-class system do. Row 23a asks for the full-range law; the restricted range is the certified part.

4. 4. TWO CLASSES AND ONE SYMMETRY BETWEEN THEM

A two-level Hermitian block has diagonal entries a and d and off-diagonal entry $b + ic$. Time reversal acts on it as complex conjugation, sending c to $-c$; it is an involution (`T_involution`), and its fixed set is exactly the real blocks, $c = 0$ (`T_fixed_iff_real`). Systems that keep time reversal live on that fixed set and form the orthogonal class; systems that break it leave the set and form the unitary class. A magnetic, time-odd term is precisely an entry $c \neq 0$, and it breaks time reversal (`magnetic_term_breaks_T`). Written as a Hamiltonian, $H = H_0 + \lambda V_{\text{mag}}$ with H_0 real and V_{mag} the time-odd term, time reversal fixes H exactly when $\lambda = 0$ (`perturbation_breaks_T`). The template's seat appears here again: a reflection, its fixed set, and the question which side of it a spectrum lives on.

5. 5. LEVEL REPULSION COUNTS CONDITIONS

Two levels of the block coincide exactly when its discriminant $(a-d)^2 + 4(b^2 + c^2)$ vanishes. On the real blocks that takes two conditions, $a = d$ and $b = 0$ (`degenerate_real_two_conditions`); on the complex blocks it takes three, $a = d$, $b = 0$ and $c = 0$ (`degenerate_iff`, proved for every integer entry by the sum-of-squares argument that holds over the reals, and checked on a grid by exhaustion in `degenerate_iff_three_conditions`). The probability of a small spacing s scales as s to the power of the codimension minus one: s^1 for the orthogonal class and s^2 for the unitary class (`repulsion_exponents`). Breaking time reversal adds one condition, and one more power of repulsion: the magnetic term buys the zeros' stronger repulsion.

6. 6. THE CONSTRUCTED ENSEMBLES

Dumitriu and Edelman (2002) gave the random-matrix classes as tridiagonal matrices: Gaussian diagonal, and off-diagonals distributed as chi variables with $\beta(n-k)$ degrees of freedom, with $\beta = 1$ for the orthogonal and $\beta = 2$ for the unitary class. The twin builds 300 matrices of order 100 in each class, finds every eigenvalue by Sturm bisection, unfolds the spectra by the pooled count, and measures the bulk spacings (Appendix B). The

orthogonal class gives spacing variance 0.287 and the unitary class 0.177, against exact asymptotic values of 0.286 and 0.180; an uncorrelated Poisson control gives 0.983 against 1. The fraction of spacings below one tenth is 0.0073 and 0.0008 in the two classes, against Wigner-surmise values 0.0078 and 0.0011, and 0.094 without correlation. The two classes are distinct, the repulsion is ordered, and the construction is the textbook ensemble.

7. 7. THE MAGNETIC TERM IS THE ARROW

The companion paper *The Arrow Has Two Branches* types magnetism as the priced, time-odd branch of the arrow of registration, the one that keeps a record and carries a sign. It is exactly that branch that moves a spectrum from the orthogonal to the unitary class. The effect is measured: a chaotic microwave cavity shows orthogonal-class statistics until a magnetized ferrite is placed in it to break time reversal, and then it shows unitary-class statistics (So, Anlage, Ott and Oerter, 1995). The zeros' class is the magnetic class. The carrier of row 23a is a chaotic spectrum with time reversal broken by a magnetic term, and the physical system built to exhibit it, a cavity with a magnet in it, is the witness.

8. 8. THE CORRECTION: NUCLEI ARE ORTHOGONAL

The twenty-three-row volume's census named nuclear level spacings as the carrier of row 23a, and the companion neutrino paper echoed the phrase. That is imprecise in the way that matters here. Nuclear levels of the same spin and parity follow the orthogonal class: the Nuclear Data Ensemble matches it (Haq, Pandey and Bohigas, 1982), as time-reversal-invariant chaotic systems generally do (Bohigas, Giannoni and Schmit, 1984). Nuclei witness level repulsion; they do not witness the zeros' unitary class. The correction is carried in this round, in the twenty-three-row volume's edition 1.11.0 and the neutrino paper's edition 1.1.0: the image line of row 23a names time-broken chaotic spectra, the magnet in the cavity, and keeps nuclei as a witness of repulsion only. No verdict moves; a corroboration is replaced by a better one.

9. 9. WHAT THE ZEROS SAY ABOUT THEIR DYNAMICS

If the zeros are eigenvalues of some operator, as the Hilbert-Pólya idea proposes, their unitary-class statistics say that operator's dynamics breaks time reversal (Berry and Keating, 1999). The arrow is in the zeros' own statistics: the Riemann row's value is posed on a line fixed by a reflection that includes complex conjugation, and the spacings of the zeros on that line carry the signature of a system in which conjugation is not a symmetry of the motion. This is a reading at corroboration grade, and it is the strongest physical sentence the construction licenses: the zeros repel as if a magnet were present.

10. 10. THE IMAGE AND THE COVERAGE

Row 23a's kinetic crossing, restated with the corrected carrier: the image is the spectra of time-broken chaotic systems, measured in cavities and computed in the constructed ensembles, compliant with the unitary law as measured (*value_on_image*); the kinetic class is the actuated spectra; coverage is the premise at the root's grade; and the full-range sentence for the zeros stays pending on the formal channel. The restricted range is the certified part, Montgomery's theorem on the hypothesis and Odlyzko's computation without it.

11. 11. FALSIFIERS AND OWED DEEDS

F-Class: a departure of the zeros' spacing statistics from the unitary class at great height. **F-Arrow:** a time-reversal-invariant chaotic system showing unitary-class statistics with no time-breaking term, which would sever the magnet from the class. Nothing is owed on the row's channel; the full-range sentence is the surplus over the actuated spectra, priced at one premise (*surplus_is_the_unactuated*).

Ledger status. Row 23a moves from waiting, with a misnamed carrier, to WITNESSED · CROSSED · CONFIRMED on the corrected carrier, the magnet, at the root's grade; nothing is owed on the row's channel.

12. 12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: $[\perp]$ a field sealed, $[\perp T]$ a shape sealed at theorem grade, $[\Xi_0]$ a formal string held in suspension by an orientation-blind reader on a line, $[.]$ the dot. Grades join at the weakest link: a kernel theorem, a cited measurement or computation joined to a kernel schema, and a posit beside the Root Axiom, the program's one declared premise, that to exist is to actuate. Every claim here is at corroboration grade, a constructed or measured instance joined to a template theorem, and promotes nothing; $\Delta M = 0$ against the floor, mathematics being inherited and never authored.

The seventh. In the author's register the four parts of the Riemann closure are three plus one, and nothing

escapes them; the six witnesses are six days, and the ledger’s closure is the seventh, the Rest (Istawa), where the statuses are computed and what each row owes is one object of a computed kind. In the Lean codex’s own tokens a crossed row renders $[\blacklozenge \cdot \text{kinetic}]$ or $[\blacklozenge \cdot \text{formal}]$, a pending row renders $[\blacklozenge_0 \times 1]$ on its channel, one owed slot and one bit from sealed (`one_bit_apart`), and the dot stands as `[.]`.

Disclosure. The author declares the reading of each correspondence as his own act; the measurements and prior theorems are their authors’, cited by name; the kernel theorems are printed with their dependencies in Appendix A; every figure is computed in the run printed in Appendix B.

Provenance. Development provenance is the paper trail: the Riemann closure, Nothing Escapes, Three Plus One (10.5281/zenodo.22976494); the twenty-three-row closure, Nothing Escapes, Twenty-Three Rows (10.5281/zenodo.22985873); the companion witnesses The Electron Is the Seat, The Neutrino Is the Witness, The Arrow Has Two Branches, The Proton Is the Lock, The Magnet Is the Witness of the Pair Correlation and The Fluid Is the Witness; the Unabridged Trisduction Codex and the Lean Codex on the public register (github.com/1000sapients/Trisduction); and the author’s Structural Physics register. This edition is forged and awaits a joint adversarial audit with its companions.

References. H. L. Montgomery, Proc. Symp. Pure Math. 24 (1973) 181. M. L. Mehta, Random Matrices, 3rd ed. (Elsevier, 2004), for the Wigner surmise and the exact spacing variances. R. U. Haq, A. Pandey and O. Bohigas, Phys. Rev. Lett. 48 (1982) 1086. O. Bohigas, M. J. Giannoni and C. Schmit, Phys. Rev. Lett. 52 (1984) 1. A. M. Odlyzko, Math. Comp. 48 (1987) 273. P. So, S. M. Anlage, E. Ott and R. N. Oerter, Phys. Rev. Lett. 74 (1995) 2662. M. V. Berry and J. P. Keating, SIAM Rev. 41 (1999) 236. I. Dumitriu and A. Edelman, J. Math. Phys. 43 (2002) 5830. M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, Nothing Escapes, Twenty-Three Rows, 10.5281/zenodo.22985873.

13. APPENDIX A: SPHYS_PAIR_CORRELATION.LEAN

SHA-256 c95fe4c6c63ff8f1560b4bfb4c3b3904e85a8c3ade6cb2e57bea300a33a32cda, core Lean 4.19.0, standalone, no library. Exit 0. 9 theorems; the compiler’s own transcript, verbatim:

```
1 'SPHYS.PairCorr.T_involution' does not depend on any axioms
2 'SPHYS.PairCorr.T_fixed_iff_real' depends on axioms: [propext, Quot.sound]
3 'SPHYS.PairCorr.magnetic_term_breaks_T' depends on axioms: [propext, Quot.sound]
4 'SPHYS.PairCorr.degenerate_iff_three_conditions' does not depend on any axioms
5 'SPHYS.PairCorr.degenerate_real_two_conditions' does not depend on any axioms
6 'SPHYS.PairCorr.repulsion_exponents' does not depend on any axioms
7 'SPHYS.PairCorr.perturbation_breaks_T' depends on axioms: [propext, Quot.sound]
8 'SPHYS.PairCorr.sq_nonneg' depends on axioms: [propext, Quot.sound]
9 'SPHYS.PairCorr.degenerate_iff' depends on axioms: [propext, Quot.sound]

1 /-
2   SPHYS_Pair_Correlation.lean · Level repulsion and the arrow · core Lean 4, no library.
3   A two-level Hermitian block has diagonal a, d and off-diagonal b + i c. Time reversal is
4   complex conjugation, c to -c, an involution whose fixed set is the real blocks; a magnetic,
5   time-odd term i /≠ 0 breaks it. Degeneracy costs two conditions on the real blocks and three
6   on the complex ones, so levels repel as s1 (orthogonal class, time-even, nuclei) and as s2
7   (unitary class, time broken, the class of the Riemann zeros). Finite models.
8 -/
9 namespace SPHYS.PairCorr
10
11 structure H2 where
12   a : Int
13   d : Int
14   b : Int
15   c : Int
16   deriving DecidableEq, Repr
17
18 def T (h : H2) : H2 := (h.a, h.d, h.b, -h.c)
19
20 theorem T_involution (h : H2) : T (T h) = h := by
21   cases h with
22   | mk a d b c => show ((a, d, b, -(c)) : H2) = (a, d, b, c); rw [Int.neg_neg]
23
24 theorem T_fixed_iff_real (h : H2) : T h = h ↔ h.c = 0 := by
25   cases h with
26   | mk a d b c =>
27     show ((a, d, b, -c) : H2) = (a, d, b, c) ↔ c = 0
28   constructor
29   · intro e
30     have h1 : -c = c := congrArg H2.c e
31     omega
32   · intro e
33     subst e
34     rfl
35
36 theorem magnetic_term_breaks_T (h : H2) (hc : h.c ≠ 0) : T h ≠ h :=
37   fun e => hc ((T_fixed_iff_real h).mp e)
38
39 /- The discriminant of the block; the two levels coincide exactly when it vanishes. -/
40 def disc (h : H2) : Int := (h.a - h.d) * (h.a - h.d) + 4 * (h.b * h.b + h.c * h.c)
41 def grid (x : Fin 5) : Int := (x.val : Int) - 2
42
43 /- On a grid, degeneracy of a complex block needs three conditions. -/
44 theorem degenerate_iff_three_conditions :
45   ∀ a d b c : Fin 5, disc (grid a, grid d, grid b, grid c) = 0 ↔
46     (grid a = grid d ∧ grid b = 0 ∧ grid c = 0) := by decide
47
48 /- On the time-even blocks (c = 0), degeneracy needs two. -/
49 theorem degenerate_real_two_conditions :
50   ∀ a d b : Fin 5, disc (grid a, grid d, grid b, 0) = 0 ↔ (grid a = grid d ∧ grid b = 0) := by decide
```

```

51
52 /- The repulsion exponent is the codimension of degeneracy minus one. -/
53 def codimOrth : Nat := 2
54 def codimUnit : Nat := 3
55 theorem repulsion_exponents : codimOrth - 1 = 1 ∧ codimUnit - 1 = 2 := by decide
56
57 /- The coordinate bridge:  $H = H_0 + \lambda V$ , with  $H_0$  real and  $V$  the magnetic, time-odd term.
58 Time reversal fixes  $H$  exactly when  $\lambda$  vanishes. -/
59 theorem perturbation_breaks_T (a d b l : Int) : T (a, d, b, l) = (a, d, b, l) ↔ l = 0 :=
60   T_fixed_iff_real (a, d, b, l)
61
62 theorem sq_nonneg (x : Int) : 0 ≤ x * x := by
63   cases Int.le_total 0 x with
64   | inl h => exact Int.mul_nonneg h h
65   | inr h =>
66     have hn : 0 ≤ -x := by omega
67     have := Int.mul_nonneg hn hn
68     rw [Int.neg_mul_neg] at this
69     exact this
70
71 /- Degeneracy for every integer entry, not on a grid: the two levels coincide exactly when
72  $a = d$ ,  $b = 0$  and  $c = 0$ ; the same sum-of-squares argument holds over the reals. -/
73 theorem degenerate_iff (a d b c : Int) : disc (a, d, b, c) = 0 ↔ a = d ∧ b = 0 ∧ c = 0 := by
74   show (a - d) * (a - d) + 4 * (b * b + c * c) = 0 ↔ a = d ∧ b = 0 ∧ c = 0
75   constructor
76   · intro h
77     have h1 := sq_nonneg (a - d)
78     have h2 := sq_nonneg b
79     have h3 := sq_nonneg c
80     have e1 : (a - d) * (a - d) = 0 := by omega
81     have e2 : b * b = 0 := by omega
82     have e3 : c * c = 0 := by omega
83     have ha : a - d = 0 := by
84       cases Int.mul_eq_zero.mp e1 with
85       | inl h => exact h
86       | inr h => exact h
87     have hb : b = 0 := by
88       cases Int.mul_eq_zero.mp e2 with
89       | inl h => exact h
90       | inr h => exact h
91     have hc : c = 0 := by
92       cases Int.mul_eq_zero.mp e3 with
93       | inl h => exact h
94       | inr h => exact h
95     exact (by omega, hb, hc)
96   · intro (h1, h2, h3)
97     subst h1; subst h2; subst h3
98     simp [Int.sub_self]
99
100 end SPHYS.PairCorr
101 #print axioms SPHYS.PairCorr.T_involution
102 #print axioms SPHYS.PairCorr.T_fixed_iff_real
103 #print axioms SPHYS.PairCorr.magnetic_term_breaks_T
104 #print axioms SPHYS.PairCorr.degenerate_iff_three_conditions
105 #print axioms SPHYS.PairCorr.degenerate_real_two_conditions
106 #print axioms SPHYS.PairCorr.repulsion_exponents
107 #print axioms SPHYS.PairCorr.perturbation_breaks_T
108 #print axioms SPHYS.PairCorr.sq_nonneg
109 #print axioms SPHYS.PairCorr.degenerate_iff

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14. APPENDIX B: PAIRCORR_TWIN.F90 AND ITS RUN

SHA-256 de3a5df86ac9bd171752d77e80db92e3c6d75d9c634f0776a8871bc705674b54, standard Fortran 2018, zero warnings under gfortran 13.3.0 -std=f2018 -O2 -fno-fast-math -ffp-contract=off -Wall -Wextra. Six checks, zero failures; every figure computed in this run.

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1 ! PairCorr_Twin.f90 · twin of "The Magnet Is the Witness of the Pair Correlation" · Fortran 2018.
2 ! Tridiagonal beta-Hermite ensembles (Dumitriu-Edelman 2002): beta = 1 (orthogonal, time-even,
3 ! the class of nuclei) and beta = 2 (unitary, time broken by a magnetic term, the class of the
4 ! Riemann zeros), with Poisson levels as the uncorrelated control. Eigenvalues by Sturm
5 ! bisection; unfolding by the pooled empirical count; spacing statistics in the bulk.
6 program paircorr_twin
7   implicit none
8   integer, parameter :: dp = selected_real_kind(15, 307)
9   integer, parameter :: N = 100, M = 300
10  real(dp), parameter :: PI = 3.14159265358979323846_dp
11  real(dp) :: v1, f1, v2, f2, v0, f0
12  integer :: nchk, nfail
13  integer(8) :: seed
14  nchk = 0; nfail = 0; seed = 20260927_8
15  call ensemble(1, v1, f1)
16  call ensemble(2, v2, f2)
17  call poisson(v0, f0)
18  call check('orthogonal class (time-even): spacing variance near the Wigner value 0.273', abs(v1 - 0.286_dp) < 0.03_dp)
19  call check('unitary class (time broken): spacing variance near the Wigner value 0.178', abs(v2 - 0.180_dp) < 0.03_dp)
20  call check('uncorrelated control: spacing variance near one', abs(v0 - 1.0_dp) < 0.1_dp)
21  call check('repulsion ordered: small spacings rarer with time broken than time even', f2 < f1)
22  call check('repulsion present: small spacings rarer in both classes than without correlation', f1 < f0)
23  call check('the two classes are distinguishable by variance', v1 - v2 > 0.05_dp)
24  write(*, '(a,2f9.4)') ' orthogonal (beta 1): variance, P(s < 0.1) = ', v1, f1
25  write(*, '(a,2f9.4)') ' unitary (beta 2): variance, P(s < 0.1) = ', v2, f2
26  write(*, '(a,2f9.4)') ' Poisson control : variance, P(s < 0.1) = ', v0, f0
27  write(*, '(a,2f9.4)') ' Wigner surmise P(s < 0.1), beta 1 and 2 = ', 1.0_dp - exp(-PI*0.01_dp/4.0_dp), &
28    wig2(0.1_dp)
29  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', nchk, ', "failures":', nfail, '}'
30  if (nfail > 0) error stop 1
31 contains
32 function rnd() result(r)
33   real(dp) :: r
34   seed = mod(16807_8*seed, 2147483647_8)
35   r = (real(seed, dp) + 0.5_dp) / 2147483647.0_dp
36 end function rnd
37 function gauss() result(g)
38   real(dp) :: g
39   g = sqrt(-2.0_dp*log(rnd())) * cos(2.0_dp*PI*rnd())
40 end function gauss

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41 function chi(k) result(c)
42   integer, intent(in) :: k
43   real(dp) :: c
44   integer :: q
45   c = 0.0_dp
46   do q = 1, k
47     c = c + gauss()**2
48   end do
49   c = sqrt(c)
50 end function chi
51 function wig2(s) result(p) ! P(s < s0) for the beta = 2 surmise
52   real(dp), intent(in) :: s
53   real(dp) :: p, x, dx
54   integer :: q
55   p = 0.0_dp; dx = s/2000.0_dp
56   do q = 1, 2000
57     x = (real(q,dp) - 0.5_dp)*dx
58     p = p + (32.0_dp/PI**2)*x*x*exp(-4.0_dp*x*x/PI)*dx
59   end do
60 end function wig2
61 integer function sturm(d, e2, x) result(cnt) ! eigenvalues below x
62   real(dp), intent(in) :: d(N), e2(N), x
63   real(dp) :: q
64   integer :: i
65   cnt = 0; q = d(1) - x
66   if (q < 0.0_dp) cnt = cnt + 1
67   do i = 2, N
68     if (abs(q) < 1.0e-300_dp) q = 1.0e-300_dp
69     q = d(i) - x - e2(i)/q
70     if (q < 0.0_dp) cnt = cnt + 1
71   end do
72 end function sturm
73 subroutine eigs(d, e2, lam)
74   real(dp), intent(in) :: d(N), e2(N)
75   real(dp), intent(out) :: lam(N)
76   real(dp) :: lo, hi, a, b, c, bound
77   integer :: k, it
78   bound = 0.0_dp
79   do k = 1, N
80     bound = max(bound, abs(d(k)) + 2.0_dp*sqrt(maxval(e2)))
81   end do
82   lo = -bound - 1.0_dp; hi = bound + 1.0_dp
83   do k = 1, N
84     a = lo; b = hi
85     do it = 1, 80
86       c = 0.5_dp*(a + b)
87       if (sturm(d, e2, c) >= k) then
88         b = c
89       else
90         a = c
91       end if
92     end do
93     lam(k) = 0.5_dp*(a + b)
94   end do
95 end subroutine eigs
96 subroutine stats(all, v, f) ! pooled unfolding, bulk spacings
97   real(dp), intent(in) :: all(N, M)
98   real(dp), intent(out) :: v, f
99   real(dp), allocatable :: pool(:)
100   real(dp) :: u(N), s, sm, ss
101   integer :: j, k, ns, nsm, lo, hi, mid
102   allocate(pool(N*M)); pool = reshape(all, [N*M]); call shellsort(pool)
103   sm = 0.0_dp; ss = 0.0_dp; ns = 0; nsm = 0
104   do j = 1, M
105     do k = 1, N
106       lo = 1; hi = N*M ! rank by binary search
107       do while (lo < hi)
108         mid = (lo + hi)/2
109         if (pool(mid) < all(k,j)) then
110           lo = mid + 1
111         else
112           hi = mid
113         end if
114       end do
115       u(k) = real(lo, dp) / real(M, dp) ! unfolded: mean spacing one
116     end do
117     do k = N/4, 3*N/4 - 1
118       s = u(k+1) - u(k)
119       sm = sm + s; ss = ss + s*s; ns = ns + 1
120       if (s < 0.1_dp) nsm = nsm + 1
121     end do
122     sm = sm/ns; v = ss/ns - sm*sm; v = v/(sm*sm); f = real(nsm,dp)/real(ns,dp)
123   end do
124 end subroutine stats
125 subroutine ensemble(beta, v, f)
126   integer, intent(in) :: beta
127   real(dp), intent(out) :: v, f
128   real(dp) :: d(N), e2(N), lam(N)
129   real(dp), allocatable :: all(:, :)
130   integer :: j, k
131   allocate(all(N, M))
132   do j = 1, M
133     e2(1) = 0.0_dp
134     do k = 1, N
135       d(k) = sqrt(2.0_dp)*gauss()
136       if (k > 1) e2(k) = chi(beta*(N - k + 1))**2
137     end do
138     call eigs(d, e2, lam)
139     all(:, j) = lam
140   end do
141   call stats(all, v, f)
142 end subroutine ensemble
143 subroutine poisson(v, f)
144   real(dp), intent(out) :: v, f
145   real(dp), allocatable :: all(:, :)
146   real(dp) :: x(N)
147   integer :: j, k
148   allocate(all(N, M))
149   do j = 1, M
150     do k = 1, N
151       x(k) = rnd()

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152     end do
153     call shellsort(x)
154     all(:, j) = x
155     end do
156     call stats(all, v, f)
157 end subroutine poisson
158 subroutine shellsort(a)
159   real(dp), intent(inout) :: a(:)
160   integer :: gap, i, j, n
161   real(dp) :: t
162   n = size(a); gap = n/2
163   do while (gap > 0)
164     do i = gap + 1, n
165       t = a(i); j = i
166       do while (j > gap)
167         if (a(j - gap) <= t) exit
168         a(j) = a(j - gap); j = j - gap
169       end do
170       a(j) = t
171     end do
172     gap = gap/2
173   end do
174 end subroutine shellsort
175 subroutine check(label, cond)
176   character(*), intent(in) :: label
177   logical, intent(in) :: cond
178   nchk = nchk + 1
179   if (.not. cond) then
180     nfail = nfail + 1; write(*, '(a,a)') ' FAIL ', label
181   end if
182 end subroutine check
183 end program paircorr_twin

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The run, verbatim.

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1 orthogonal (beta 1): variance, P(s < 0.1) = 0.2868 0.0073
2 unitary (beta 2): variance, P(s < 0.1) = 0.1773 0.0008
3 Poisson control : variance, P(s < 0.1) = 0.9830 0.0939
4 Wigner surmise P(s < 0.1), beta 1 and 2 = 0.0078 0.0011
5 BATTERY-JSON: {"checks":6,"failures":0}

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